

Integration with Server

android 

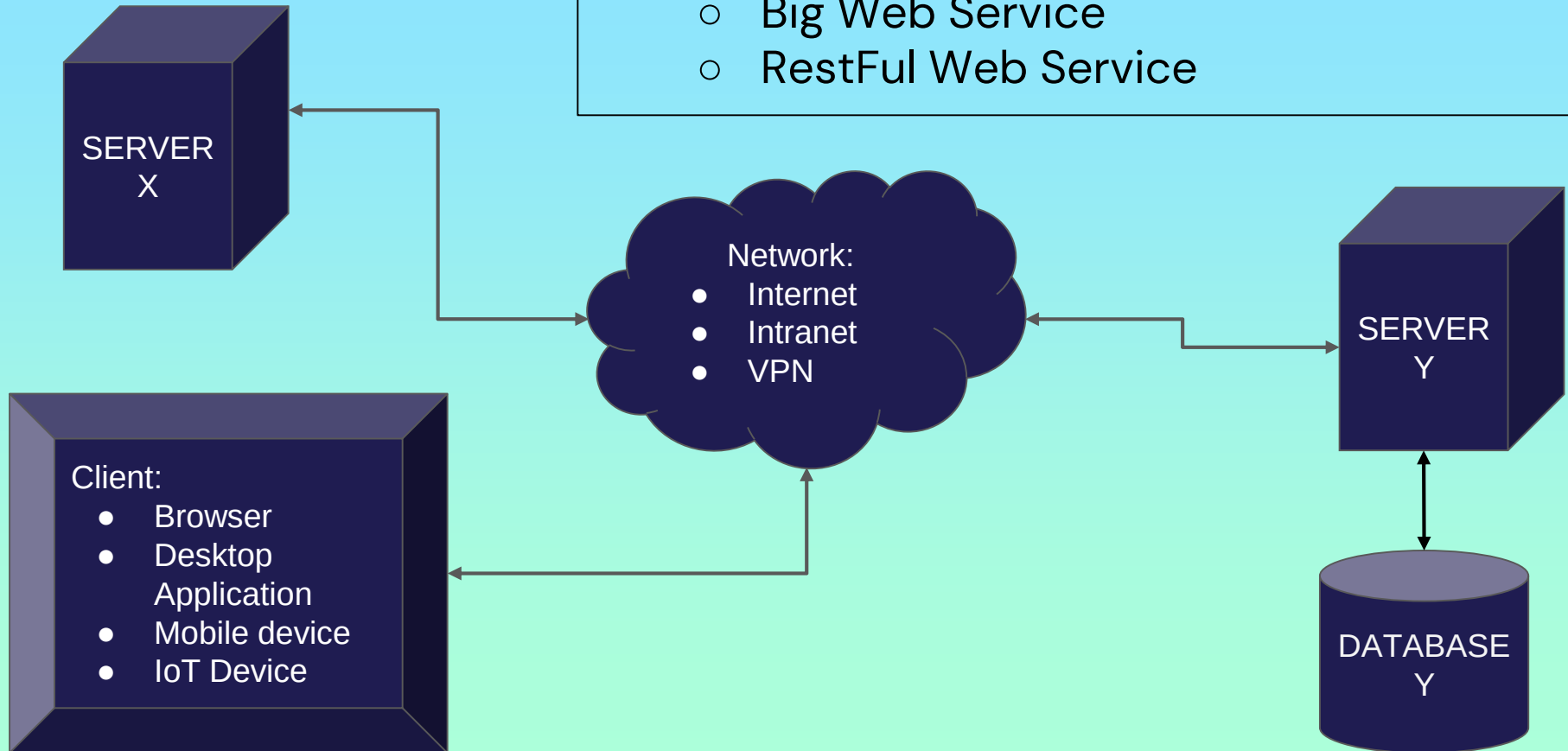
Web Service Overview

android 

Web Service

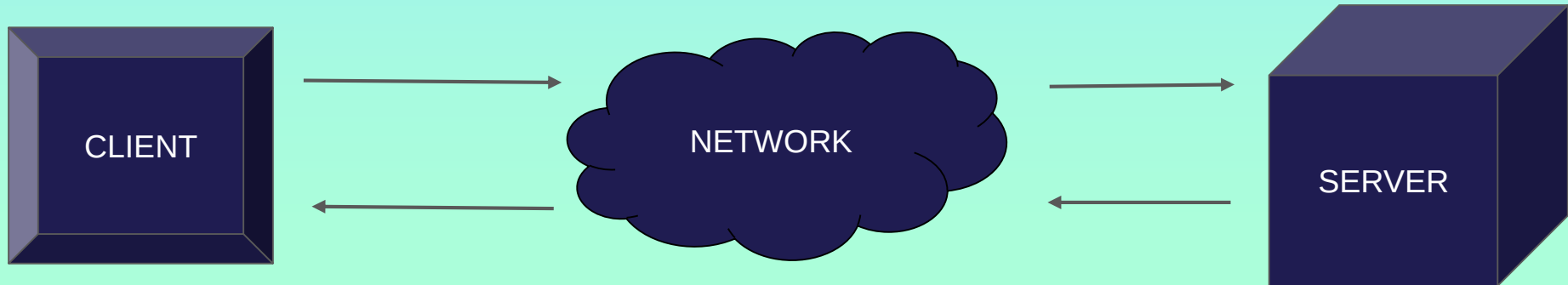
Komponen software yang di akses melalui jaringan komputer

- Big Web Service
- RestFul Web Service



RestFul Web Service

- Protocol : HTTP/s
- Method :
 - GET → Baca data
 - POST → Tambah data
 - PUT → Ubah data
 - DELETE → Hapus data
- Data Format : JSON/XML



JSON

android 

Json Object

- Dalam kurung kurawal { ... }
- Item Data adalah pasangan **Key : Value**
- Tipe data **key** → **String/Text**
- Tipe data valid **value**:
 - String
 - Number
 - JSON Objek
 - Array
 - Boolean
 - null

```
{
  "id": 1,
  "name": "Leanne Graham",
  "username": "Bret",
  "email": "Sincere@april.biz",
  "address": {
    "street": "Kulas Light",
    "suite": "Apt. 556",
    "city": "Gwenborough",
    "zipcode": "92998-3874",
    "geo": {
      "lat": "-37.3159",
      "lng": "81.1496"
    }
  },
  "phone": "1-770-736-8031 x56442",
  "website": "hildegard.org",
  "company": {
    "name": "Romaguera-Crona",
    "catchPhrase": "Multi-layered client-server neural-net",
    "bs": "harness real-time e-markets"
  }
}
```

Json Array

- Dalam kurung Siku [...]
- Setiap item → Tipe data valid value

```
[
  {
    "postId": 1,
    "id": 1,
    "name": "id labore ex et quam laborum",
    "email": "Eliseo@gardner.biz",
    "body": "laudantium enim quasi est quidem magnam voluptate ipsam eos\ntempora quo necessitatibus\ndolor quam autem quasi\nreiciendis et nam sapiente accusantium"
  },
  {
    "postId": 1,
    "id": 2,
    "name": "quo vero reiciendis velit similique earum",
    "email": "Jayne_Kuhic@sydney.com",
    "body": "est natus enim nihil est dolore omnis voluptatem numquam\net omnis occaecati quod ullam at\nvoluptatem error expedita pariatur\nnnihil sint nostrum voluptatem reiciendis et"
  },
  {
    "postId": 1,
    "id": 3,
    "name": "odio adipisci rerum aut animi",
    "email": "Nikita@garfield.biz",
    "body": "quia molestiae reprehenderit quasi aspernatur\naut expedita occaecati aliquam eveniet laudantium\nomnis quibusdam delectus saepe quia accusamus maiores nam est\ncum et ducimus et vero voluptates excepturi deleniti ratione"
  },
  {
    "postId": 1,
    "id": 4,
    "name": "alias odio sit",
    "email": "Lew@alysa.tv",
    "body": "non et atque\noccaecati deserunt quas accusantium unde odit nobis qui voluptatem\nquia voluptas consequuntur itaque dolor\net qui rerum deleniti ut occaecati"
  },
  {
    "postId": 1,
    "id": 5,
    "name": "vero eaque aliquid doloribus et culpa",
    "email": "Hayden@althea.biz",
    "body": "harum non quasi et ratione\ntempore iure ex voluptates in ratione\nharum architecto fugit inventore cupiditate\nvoluptates magni quo et"
  }
]
```

JSON & XML Parsing

android 

Parse json - <https://www.jsonschema2pojo.org/>

Paste objek/array json pada textbox

Generate Plain Old Java Objects from JSON or JSON-Schema.

```
1 {  
2   "userId": 1,  
3   "id": 1,  
4   "title": "sunt aut facere repellat provident occaecati excepturi  
5   "body": "quia et suscipit\nsuscipit recusandae consequuntur expedit  
6 }
```

Package

Class name

Source type:

☒ JSON Schema ☐ JSON

☐ YAML Schema ☐ YAML

Annotation style:

☐ Jackson 2.x ☒ Gson

☐ Moshi ☐ None

☐ Generate builder methods

☐ Use primitive types

☐ Use long integers

☒ Use double numbers

☐ Use Joda dates

☒ Include getters and setters

☐ Include constructors

☐ Include `hashCode` and `equals`

☐ Include `toString`

☐ Include JSR-303 annotations

☒ Allow additional properties

☐ Make classes serializable

☐ Make classes parcelable

☐ Initialize collections

Property word delimiters:

Sesuaikan nama package dan kelas

Sesuaikan sumber data (JSON)

Sesuaikan tipe anotasi / library parser (Gson)

Info lainnya default

Klik untuk preview kelas parser

Preview

Zip

Retrofit

android 

Retrofit - Networking Library

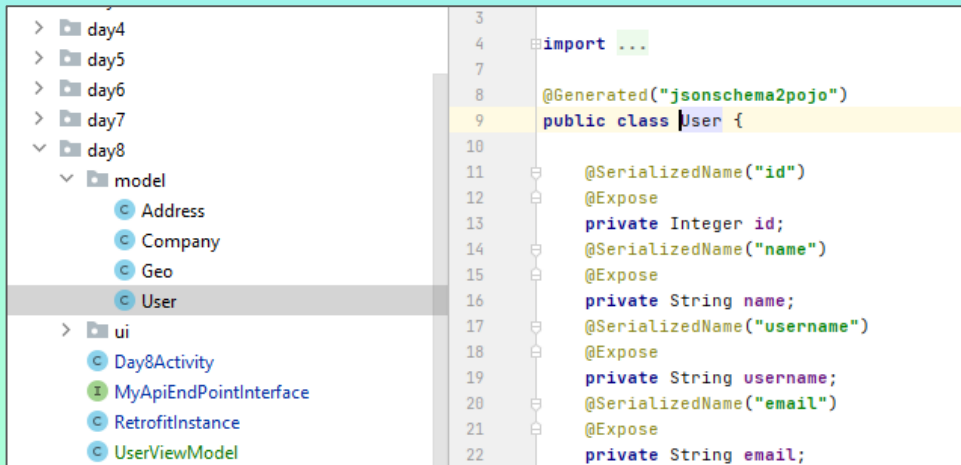
1. Tambahkan permission internet di manifest

```
<uses-permission android:name="android.permission.INTERNET" />
```

2. Tambahkan dependencies di build.gradle module

```
implementation 'com.google.code.gson:gson:2.8.7'  
implementation 'com.squareup.retrofit2:retrofit:2.9.0'  
implementation 'com.squareup.retrofit2:converter-gson:2.9.0'  
implementation 'org.glassfish:javax.annotation:10.0-b28'
```

3. Tambahkan kelas model



4. Definiskan endpoint dan method http confignya

```
public interface MyApiEndPointInterface {  
  
    // Trailing slash is needed  
    String BASE_URL = "https://jsonplaceholder.typicode.com";  
  
    @GET("/users")  
    Call<List<User>> getUsers();  
  
    @GET("/users/{id}")  
    Call<User> getUser(@Path("id") String id);  
}
```

5. Definiskan kelas Retrofit beserta confignya

```
public class RetrofitInstance {  
    private MyApiEndPointInterface API;  
  
    public RetrofitInstance(){  
        Gson gson = new GsonBuilder()  
            .setDateFormat("yyyy-MM-dd'T'HH:mm:ssZ")  
            .create();  
        Retrofit retrofit = new Retrofit.Builder()  
            .baseUrl(MyApiEndPointInterface.BASE_URL)  
            .addConverterFactory(GsonConverterFactory.create(gson))  
            .build();  
        retrofit.create(MyApiEndPointInterface.class);  
        API = retrofit.create(MyApiEndPointInterface.class);  
    }  
  
    public MyApiEndPointInterface getAPI(){  
        return API;  
    }  
}
```

Retrofit - Networking Library

7. Definiskan callback callable dengan meng override method onResponse dan onFailure Pada fragment

```
@Override
public void onViewCreated(@NonNull View view, @Nullable Bundle savedInstanceState) {
    super.onViewCreated(view, savedInstanceState);
    userModel
        .getRetrofitInstance() . RetrofitInstance
        .getAPI() . MyApiEndPointInterface
        .getUsers() . Call<List<User>>
        .enqueue(new Callback<List<User>>() {
            @Override
            public void onResponse(Call<List<User>> call, Response<List<User>> response) {
                pb.setVisibility(View.INVISIBLE);
                userAdapter.submitList(response.body());
            }
            @Override
            public void onFailure(Call<List<User>> call, Throwable t) {
                pb.setVisibility(View.INVISIBLE);
                Snackbar.make(view, "Error : " + t.getMessage(), Snackbar.LENGTH_INDEFINITE).show();
            }
        });
}
```

6. Instansiasi retrofit instance dalam view model

```
public class UserViewModel extends AndroidViewModel {

    private RetrofitInstance retrofitInstance;

    public UserViewModel(@NonNull Application application) {
        super(application);
        retrofitInstance = new RetrofitInstance();
    }

    public RetrofitInstance getRetrofitInstance(){
        return retrofitInstance;
    }
}
```

Retrofit - Parameter

Anotasi	Deskripsi
@Path	Substitusi variabel untuk Endpoints API (yaitu nama pengguna akan ditukar dengan {username} di Endpoint URL).
@Query	Menentukan pasangan kunci dan value untuk kueri URL parameter.
@Body	Data pada body http (objek Java atau string JSON)
@Header	Konfigurasi pada header http

Codebase

- <https://github.com/ipaddr/BasicAndroid>



THANK
YOU